

Network Address Translation (NAT)

To go to the Internet we need to get an public IP address and it is unique all over the world. If each host in the world required a unique public IP address, we would have run out of IP addresses years ago. But by using Network Address Translation (NAT) we can save tons of IP addresses for later uses. We can understand NAT like this:

“NAT allows a host that does not have a valid registered IP address to communicate with other hosts through the Internet”

For example your computer is assigned a private IP address of 10.0.0.9 and of course this address can not be routed on the internet but you can still access the internet. This is because your router (or modem) translates this address into a public IP address, 123.12.23.1 for example, before routing your data into the internet.



Of course when your router receives a reply packet destined for 123.12.23.1 it will convert back to your private IP 10.0.0.9 before sending that packet to you.

Maybe you will ask “hey, I don’t see any difference of using NAT to save tons of IP addresses because you still need a public IP address for each host to access the Internet and it doesn’t save you anything, why you need to use NAT?”

Ok, you are right :), in the above example we don’t see its usefulness but you now understand the fundamental of NAT!

Let’s take another example!

Suppose your company has 500 employees but your Internet Service Provider (ISP) only gives you 50 public IP addresses. It means that you can only allow 50 hosts to access the internet at the same time. Here NAT comes to save your life!

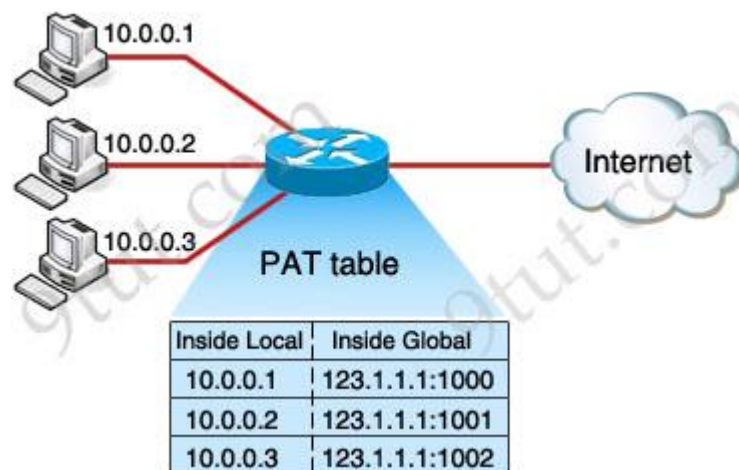
One thing you should notice that in real life, not all of your employees uses internet at the same time. Say, maybe 50 of them use internet to read newspaper at the morning; 50 others use internet at noon for checking mail... By using NAT you can dynamically assign these 50 public IP addresses to those who really need them at that time. This is called **dynamic NAT**.

But the above NAT solution does not solve our problem completely because in some days there can be more than 50 people surfing web at the morning. In this case, only the first 50 people can access internet, others must wait to their turns.

Another problem is, in fact, your ISP only gives you much lesser IP addresses than the number 50 because each public IP is very precious now.

To solve the two problems above, another feature of NAT can be used: **NAT Overload** or sometimes called **Port Address Translation (PAT)**

PAT permits multiple devices on a local area network (LAN) to be mapped to a single public IP address with different port numbers. Therefore, it's also known as port address translation (PAT). When using PAT, the router maintains unique source port numbers on the **inside global** IP address to distinguish between translations. In the below example, each host is assigned to the same public IP address 123.1.1.1 but with different port numbers (from 1000 to 1002).



Note: Cisco uses the term **inside local** for the private IP addresses and **inside global** for the public IP addresses replaced by the router.

The outside host IP address can also be changed with NAT. The **outside global** address represents the outside host with a public IP address that can be used for routing in the public Internet.

The last term, **outside local** address, is a private address of an external device as it is referred to by devices on its local network. You can understand outside local address as the inside local address of the external device which lies at the other end of the Internet.

Maybe you will ask how many ports can we use for each IP? Well, because the port number field has 16 bits, PAT can support about 2^{16} ports, which is more than 64,000 connections using one public IP address.

Now you have learned all the most useful features of NAT but we should summary all features of NAT:

There are two types of NAT translation: dynamic and static.

Static NAT: Designed to allow one-to-one mapping between local and global addresses. This flavour requires you to have one real Internet IP address for every host on your network.

Dynamic NAT: Designed to map an unregistered IP address to a registered IP address from a pool of registered IP addresses. You don't have to statically configure your router to map an

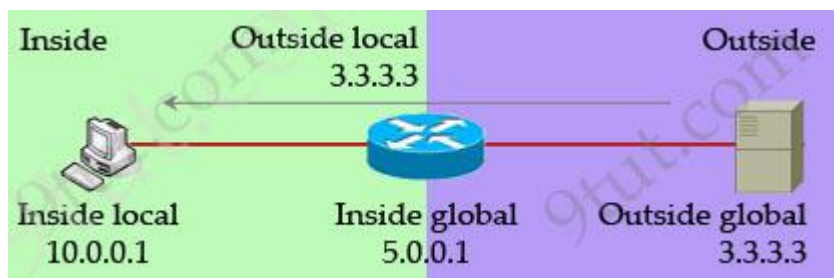
inside to an outside address as in static NAT, but you do have to have enough real IP addresses for everyone who wants to send packets through the Internet. With dynamic NAT, you can configure the NAT router with more IP addresses in the inside local address list than in the inside global address pool. When being defined in the inside global address pool, the router allocates registered public IP addresses from the pool until all are allocated. If all the public IP addresses are already allocated, the router discards the packet that requires a public IP address.

PAT (NAT Overloading): is also a kind of dynamic NAT that maps multiple private IP addresses to a single public IP address (many-to-one) by using different ports. Static NAT and Dynamic NAT both require a one-to-one mapping from the inside local to the inside global address. By using PAT, you can have thousands of users connect to the Internet using only one real global IP address. PAT is the technology that helps us not run out of public IP address on the Internet. This is the most popular type of NAT.

Besides NAT gives you the option to advertise only a single address for your entire network to the outside world. Doing this effectively hides the internal network from the public world really well, giving you some additional security for your network.

NAT terms:

- * **Inside local address** – The IP address assigned to a host on the inside network. The address is usually not an IP address assigned by the Internet Network Information Center (InterNIC) or service provider. This address is likely to be an RFC 1918 private address.
- * **Inside global address** – A legitimate IP address assigned by the InterNIC or service provider that represents one or more inside local IP addresses to the outside world.
- * **Outside local address** – The IP address of an outside host as it is known to the hosts on the inside network.
- * **Outside global address** – The IP address assigned to a host on the outside network. The owner of the host assigns this address.



To learn how to configure NAT please read my [Configure NAT GNS3 Lab tutorial](#)

Example of Static NAT :

Configuring Static NAT Using Cisco IOS

NAT enables hosts with internal private addresses to communicate on the Internet. When configuring NAT, at least one interface must be configured as the inside interface. The inside interface is connected to the internal, private network. Another interface, usually the external interface used to access the Internet, must be configured as the outside interface. When devices on the internal network communicate out through the external interface, the addresses are translated to one or more registered IP addresses.

There are occasions when a server located on an internal network must be accessible from the Internet. This accessibility requires that the server has a specific registered address that external users can specify. One way to provide this address to an internal server is to configure a static translation.

Static NAT ensures that addresses assigned to hosts on the internal network are always translated to the same registered IP address.

```
Router(config)# interface fastethernet 0/0
Router(config-if)# ip address 172.31.232.182 255.255.255.0
Router(config-if)# ip nat inside
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface serial 0/0
Router(config-if)# ip address 209.165.201.1 255.255.255.252
Router(config-if)# ip nat outside
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# ip nat inside source static 172.31.232.14 209.165.202.130
Router(config)# exit

show running-config
```

Configuring NAT and static NAT using the Cisco IOS CLI requires a number of steps.

- **Step 1.** Specify the inside interface.
- **Step 2.** Set the primary IP address of the inside interface.
- **Step 3.** Identify the inside interface using the ip nat inside command.
- **Step 4.** Specify the outside interface.

- **Step 5.** Set the primary IP address of the outside interface.
- **Step 6.** Identify the outside interface using the ip nat outside command.
- **Step 7.** Define the static address translation.
- **Step 8.** Verify the configuration.

There are several router CLI commands to view NAT operations for verification and troubleshooting.

One of the most useful commands is **show ip nat translations**. The output displays the detailed NAT assignments. The command shows all static translations that have been configured and any dynamic translations that have been created by traffic. Each translation is identified by protocol and its inside and outside local and global addresses.

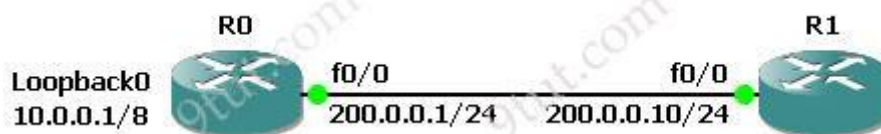
The **show ip nat statistics** command displays information about the total number of active translations, NAT configuration parameters, how many addresses are in the pool, and how many have been allocated.

Additionally, use the **show run** command to view NAT configurations.

Another Example of Static NAT

R0router(config-if)#ip nat inside (or ip nat outside)

R0router(config)#ip nat inside source static



Therefore we should configure the loopback interface of R0 as the source IP address and the fa0/0 interface of R0 as the "outgoing static NAT" address.

R0#configure terminal

R0(config)#int loopback0

R0(config-if)#ip address 10.0.0.1 255.0.0.0

R0(config-if)#ip nat inside

R0(config-if)#int f0/0

R0(config-if)#ip address 200.0.0.1 255.255.255.0

R0(config-if)#no shutdown

R0(config-if)#ip nat outside

R0(config-if)#exit

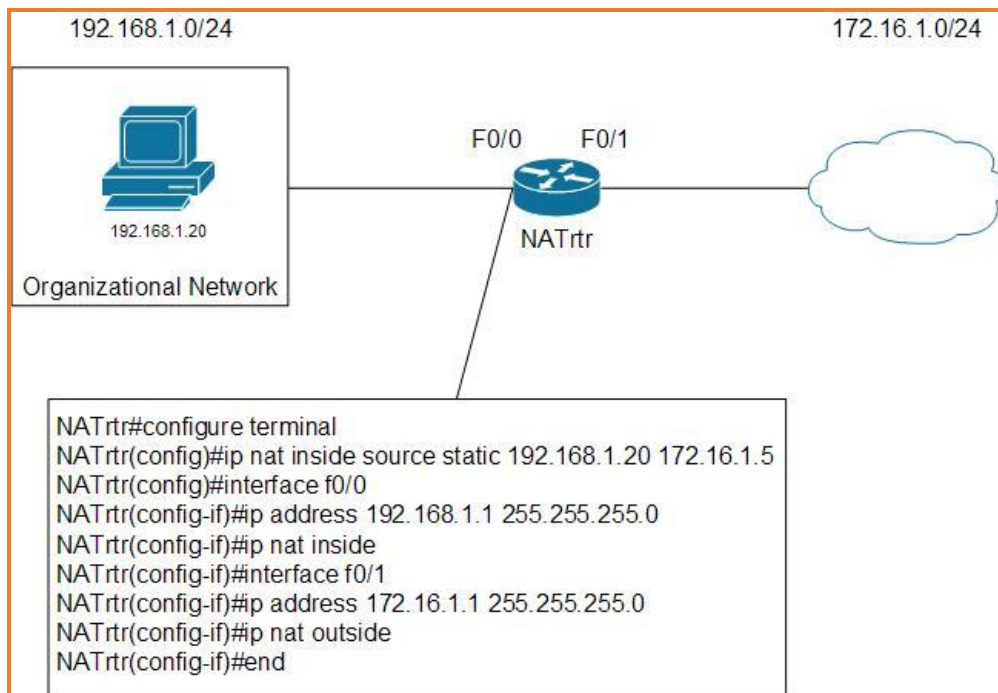
Finally, we have to tell the router to translate my private IP **10.0.0.1** to public IP **200.0.0.2** so that I can go to the Internet!

R0(config)#ip nat inside source static 10.0.0.1 200.0.0.2

In R1 we just assign the IP address and no shut its interface.

```
R1#config terminal
R1(config)#int f0/0
R1(config-if)#ip address 200.0.0.10 255.255.255.0
R1(config-if)#no shutdown
```

Another Example of Static NAT :



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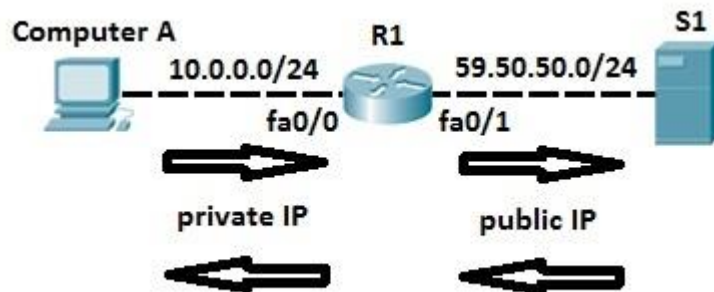
With static NAT, routers or firewalls translate one private IP address to one public IP address. Each private IP address has been mapped to a single public IP address. Static NAT is not often used because it requires one public IP address for each private IP address.

To configure static NAT, three steps are required:

1. configure private/public IP address mapping by using the *ip nat inside source static PRIVATE_IP PUBLIC_IP* command
2. configure the router's inside interface using the *ip nat inside* command

3. configure the router's outside interface using the *ip nat outside* command

Here is an example.



Computer A requests a web resource from S1. Computer A uses its private IP address when sending the request to router R1. Router R1 receives the request, changes the private IP address to the public one and sends the request to S1. S1 responds to R1. R1 receives the response, looks up in its NAT table and changes the destination IP address to the private IP address of Computer A.

In the example above, we need to configure static NAT. To do that, the following commands are required on R1:

```
R1(config)#ip nat inside source static 10.0.0.2 59.50.50.1
R1(config)#int fa0/0
R1(config-if)#ip nat inside
R1(config-if)#int fa0/1
R1(config-if)#ip nat outside
```